

Tech & Teach

Digital Chip Design Program

RTL Design

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Task #07

15/7/2026

Sequential Datapath Design Part 1

- Blocking & NonBlocking

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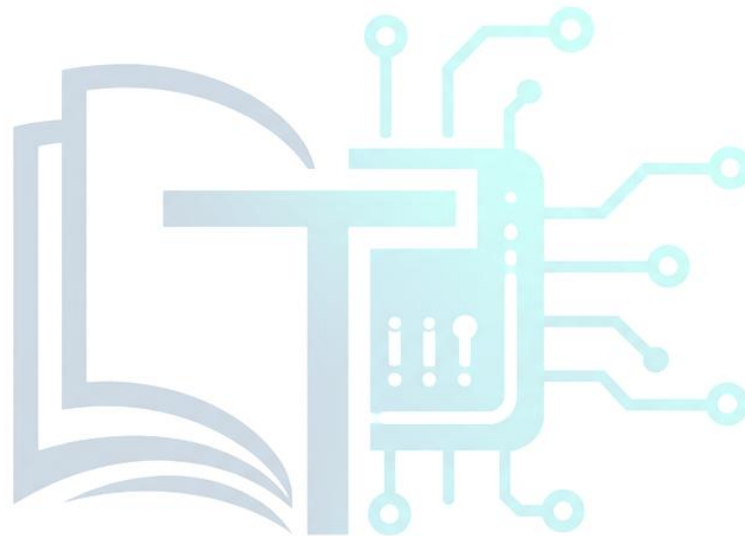
Task #1.1.1

```
module pipo_register (  
    input logic    clk,  
    input logic    reset,  
    input logic [3:0] D,  
    output logic [3:0] Q  
);  
  
    always_ff @(posedge clk or posedge reset) begin  
        if (reset) begin  
            Q <= 4'b0000;  
        end else begin  
            Q <= D;  
        end  
    end  
  
endmodule
```

Task #1.1.2

```
`timescale 1ns / 1ps  
  
module pipo_register_tb();  
  
    logic    clk;  
    logic    reset;  
    logic [3:0] D;  
    logic [3:0] Q;  
  
    pipo_register dut (  
        .clk(clk),  
        .reset(reset),  
        .D(D),  
        .Q(Q)  
    );  
  
    always #5 clk = ~clk;  
  
    initial begin  
        clk = 0;  
        reset = 1;  
        D = 4'b0000;  
        #10;  
  
        reset = 0;
```

```
#10 D = 4'b1010;  
#10 D = 4'b0110;  
#10 D = 4'b1111;  
#10 D = 4'b0000;  
  
#10 $finish;  
end  
  
endmodule
```



Task #1.2.1

```
module siso_shift_reg (  
    input logic clk,  
    input logic reset,  
    input logic SI,  
    output logic SO  
);  
  
    logic [3:0] shift_reg;  
  
    always_ff @(posedge clk or posedge reset) begin  
        if (reset) begin  
            shift_reg <= 4'b0000;  
        end else begin  
            shift_reg <= {SI, shift_reg[3:1]};  
        end  
    end
```

```
end

    assign SO = shift_reg[0];

endmodule
```

Task #1.2.2

```
`timescale 1ns / 1ps

module siso_shift_reg_tb();

    logic clk;
    logic reset;
    logic SI;
    logic SO;

    siso_shift_reg dut (
        .clk(clk),
        .reset(reset),
        .SI(SI),
        .SO(SO)
    );

    always #5 clk = ~clk;

    initial begin
        clk = 0;
        reset = 1;
        SI = 0;
        #10;

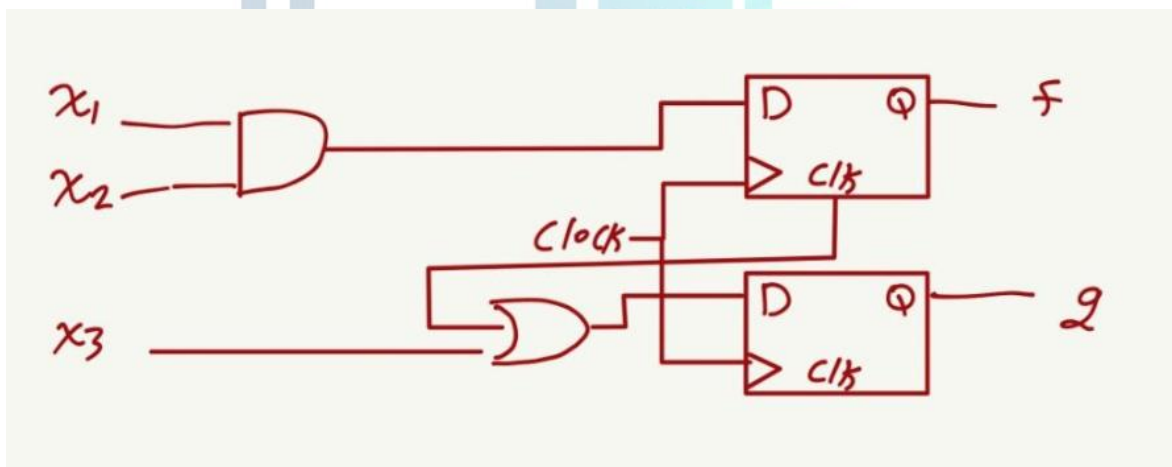
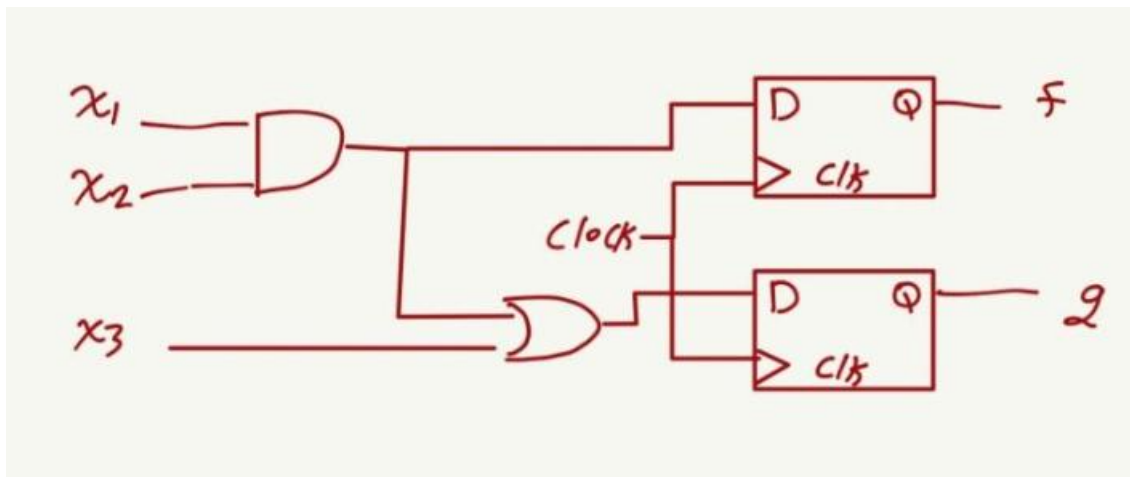
        reset = 0;

        #10 SI = 1;
        #10 SI = 0;
        #10 SI = 1;
        #10 SI = 1;

        #40;
        $finish;
    end

endmodule
```

Task #3.1



Work distribution

Section	Abdulmalek	Omar	Khaled
Report	✓		
Task n.x		✓	
Task n.x+1			✓

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